

Harshita Karnam

Engineering Undergraduate

Second-year B.Tech CSE student at Indian Institute of Information Technology Sri City with interests in AI and intelligent traffic systems. Published research (under review) on LLM-guided RL for adaptive traffic control. Strong problem-solving, teamwork, and communication skills.

Tel: +91 8125487298 **Email:** harshita.karnam909@gmail.com
LinkedIn: [harshitakarnam7b6072318](https://www.linkedin.com/in/harshitakarnam7b6072318)
GitHub: github.com/harshitak13
Portfolio Link: portfolio.vercel.app



EDUCATION

B.Tech — Computer Science & Engineering *2024 – 2028 (Completed 2nd Year)*
Indian Institute of Information Technology, Sri City, India

TECHNICAL SKILLS

Languages	Python, C, C++, HTML, CSS, JavaScript
ML / RL	PyTorch, OpenAI Gym, SUMO
Algorithms	DQN, Double DQN, PPO, CoLight, Multi-Agent RL
Concepts	Graph Attention Networks, LLM Integration, Reward Shaping, Policy Gradient
Tools	Git / GitHub, Linux, VS Code

RESEARCH

SafeGAT (*Manuscript Under Review*) — LLM-augmented Graph Attention RL framework for adaptive traffic signal control; outperforms pure RL baselines on unseen road topologies.

PROJECTS

SafeGAT — LLM-Integrated RL Traffic Signal Control

- Hybrid architecture coupling LLMs with GAT-based RL for adaptive traffic control; improved generalisation across unseen intersections via LLM-derived auxiliary policy guidance.
- Manuscript under peer review; second-year undergraduate work.
- [GitHub: harshitak13/SafeGat](https://github.com/harshitak13/SafeGat)

RL-Based Traffic Signal Control (DQN / DDQN / PPO / CoLight)

- Benchmarked four RL algorithms across single and 4x4 multi-intersection grids; DDQN and CoLight led on reward, queue length, and travel time.
- [GitHub: harshitak13/RL-TSC](https://github.com/harshitak13/RL-TSC)

DQN CartPole — Deep Q-Network from Scratch

- Built a full DQN agent with experience replay and target network on CartPole-v1; ablation study vs. vanilla Q-learning validated replay buffer and target-network stabilisation.
- [GitHub: harshitak13/DQN_Cartpole](https://github.com/harshitak13/DQN_Cartpole)

EXPERIENCE

Director, Management Dept. — Entrepreneurship Cell, IIIT Sri City *Sept 2025 – Present*

- Organised AI/ML workshops and tech events; led cross-functional student teams.

Member, Developer Relations — Google Developer Groups (GDG) *Sept 2025 – Present*

- Participated in developer events, study jams, and collaborative ML sessions.